

16. (1.75 points)

Given the following reinsurance company data (\$000) as of December 31, 2015:

Calendar-Accident Year	Earned Risk Pure Premium	Adjusted Premium	Aggregate Reported Loss	Reported Loss Development Factor to Ultimate
2012	20,500	21,000	9,000	1.250
2013	21,500	27,000	6,000	2.500
2014	22,800	25,000	4,000	5.000
<u>2015</u>	<u>24,000</u>	<u>24,000</u>	<u>3,250</u>	8.000
Total	88,800	97,000	22,250	n/a

- The selected credibility factor is 0.80.

a. (1 point)

Use the Stanard-Bühlmann method to estimate the IBNR for all years combined.

b. (0.75 point)

Use the credibility weighting of the Stanard-Bühlmann and chain ladder methods as presented by Patrik to estimate the IBNR for accident year 2014.

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EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 16	
TOTAL POINT VALUE: 1.75	LEARNING OBJECTIVE: A12: Adjust primary methods and data to be used for reinsurance reserving.
SAMPLE ANSWERS	
Part a: 1.0 point	
<i>Sample Answer 1</i> $SB\ ELR = \sum ARL(k) / \sum ARPP(K) Rlag(k) = 22,500 / 35,600 = 0.625$ $Rlag = 1/LDF$ $SB\ IBNR = \sum ARPP(k)(1 - Rlag(k)) ELR = 38,375,000$ <i>Sample Answer 2</i> $ELR = 22,250 / \text{reported } \times \text{Adj prem} = 22,250 / \left(21,000 \left(\frac{1}{1.25} \right) + \dots + 24,000 \left(\frac{1}{8} \right) \right) = 0.625$ $IBNR = 21,000 (0.625) (1 - 1/1.25) + \dots + 24,000 (0.625) (1 - 1/8) = 38,375$	
Part b: 0.75 point	
<i>Sample Answer 1</i> $P_{2014} = 0.2$ $Z = 0.2 \times 0.8 = 0.16$ $R_{CL} = 4,000 / 0.2 \times 0.8 = 16,000$ $R_{SB} = 25 \times 0.625 \times 0.8 = 12,500$ $IBNR_{cred} = 0.16 (16,000) + (1 - 0.16) (12,500) = 13,060$ <i>Sample Answer 2</i> $CL\ res = Rep \times LDF_{ult} - Rep$ 2014: $4,000 \times 5 - 4,000 = 16,000$ $Z = p \times cred = 0.2 \times 0.8 = 0.16$ $Wtd\ res = z (CL) + (1 - z) SB$ $(0.16) (16,000) + (1 - 0.16) (12,500) = 13,060,000$	
EXAMINER'S REPORT	
Part a	
Candidates were expected to execute the mechanics of the Stanard-Bühlmann method to estimate IBNR. Generally, candidates performed very well and most candidates earned full credit or lost credit only for a computation error. The most common error was to use 22,500 for reported losses instead of the given 22,250. A limited number of candidates also used Earned-Risk Pure Premium data instead of the Adjusted Premium data to calculate the unused premium in the last step of the calculation.	
Part b	
Candidates were expected to know Patrik's approach to credibility-weighting the IBNR estimated in part a. against a chain-ladder IBNR estimate. Generally, candidates performed well and most candidates earned full credit or lost credit only for a computation error. The most common errors were:	

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

- Calculating IBNR in total for the 4 AY data and not isolating AY2014 or using a wrong AY (e.g., AY2015)
- Not calculating the credibility weights properly. Many candidates used $Z=0.8$ directly as provided in the question instead of applying the report-lag factor to it.
- Using a wrong ultimate loss number for the SB component. Some candidates applied the ELR to the adjusted premium and subtracted reported losses.
- Using ultimate number for CL method as if it is an IBNR number (i.e., not deducting 1.0 from the LDFs).